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### Tidal Hiccups and Oxidic Sighs: Machine-Learning classification of oxygenation events in a transitioning coastal wetland

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#### Abstract Text:

Coastal wetlands are converting former freshwater floodplains into tidally influenced marsh, reshaping the subsurface dissolved-oxygen (DO) regime that governs biogeochemical function. At Beaver Creek, Washington, a near-anoxic floodplain well (DO levels below detection  $\sim 92\%$  of the time), punctuated by episodic oxygenation events. These events fall into two physically distinct classes: abrupt, asymmetric "hot moments" driven by tidal/saline intrusion, and slower, symmetric "oxidic pulses" driven by freshwater rise from precipitation/flooding. We formalize separating these regimes as a supervised classification problem using six years of co-located 5-minute DO, hydrology (water level, salinity, temperature), and weather measurements.

We compare two modeling tracks. The first uses tabular classification models on domain-engineered event-shape features encoding the morphology of the DO vs. time data, change in salinity and water level, antecedent-precipitation, etc. The second directly uses raw multivariate time-series data either as inputs to a Convolutional Neural Network (CNN) or through automatic encoding into  $\mathcal{O}(10^4)$  features capturing correlations over various time scales, fed into tabular models. Classification performance is reported through metrics like Confusion Matrix, F1-score, Balanced Accuracy, etc.

Furthermore, through feature importance attribution, we are isolating and identifying the various drivers for these DO events.

**Session Selection:**

B017. Advancing Wetland Biogeochemical and Ecohydrological Processes:  
Observations, Models, and Emerging Approaches

**Submitter's E-mail Address:**

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**Abstract Title:**

Tidal Hiccups and Oxidic Sighs: Machine-Learning classification of oxygenation  
events in a transitioning coastal wetland

**Requested Presentation Type:**

Poster Only

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Yes

**Previously Published?:**

No

**Abstract Payment:**

Paid (agu-agu26-2085144-5407-6635-5947-6995)

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